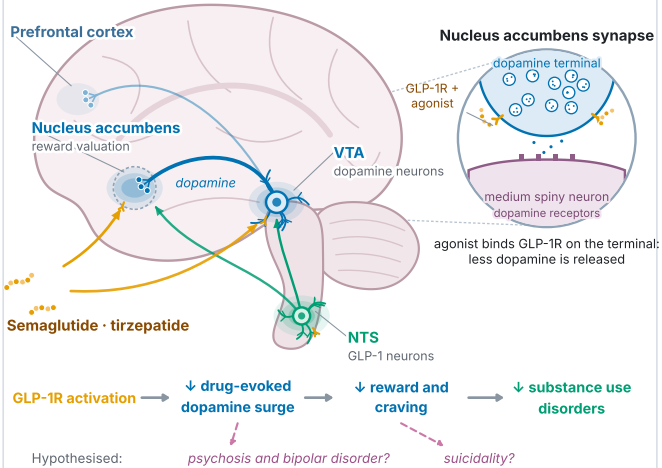
Semaglutide, tirzepatide and the psychiatric phenome

Primary analysis plus 8 secondary target trials in TriNetX · new users · 19 psychiatric outcomes over 3 years · calibrated to 6 negative controls

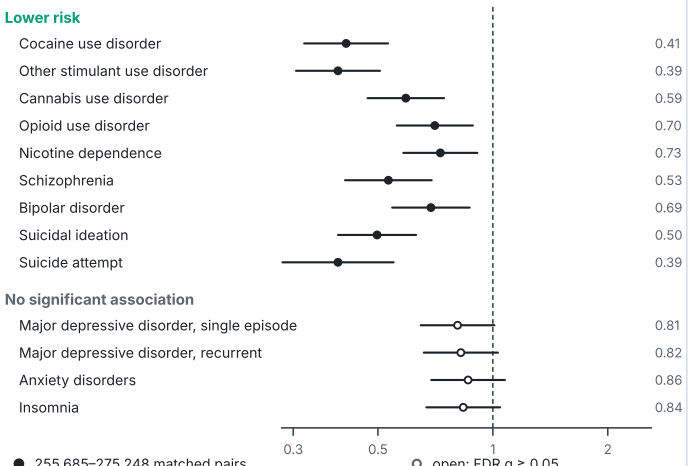
Proposed mechanism

GLP-1 signalling in the mesolimbic dopamine reward pathway (mainly rodent studies)



Primary analysis, type 2 diabetes

Semaglutide or tirzepatide vs SGLT2 inhibitor, 2022–2025: calibrated HR (95% CI)



Findings replicated for each drug separately, against DPP-4 inhibitors and untreated patients, and partly in obesity; tirzepatide and semaglutide did not differ significantly.

Observational associations: randomized trials in stimulant use disorder, early psychosis and suicide prevention are warranted.